CANDY
DIALOGUE ENGINE

Speech Bubbles

1.1.0

Candy DE features the option to display spoken text inside speech bubbles, like in comic books.

This requires a tiny bit of setup, but nothing complicated.

Setup

Candy DE provides two default speech bubble scenes: one for 2D environments and one for 3D environments. These scenes are located in the **Candy_DE/Scenes/Speech Bubbles 2D** and the **Candy_DE/Scenes/Speech Bubbles 3D** folders.

Bubble Locations

Those speech bubble scenes need to be added to characters, anywhere you want inside their node hierarchy. The root node of your characters must have a script, inside of which you must provide the path to the speech bubble, exactly as shown below:

```
@onready var internal_node_dict = {
	"Speech_Bubble": get_node("SpeechBubble"),
}
```

For easy copy-paste:

```
@onready var internal_node_dict = {
      "Speech_Bubble": get_node("SpeechBubble"),
}
```

This will allow Candy DE to locate speech bubbles, regardless of how you setup the node hierarchy of characters.

Character Locations

Next, you must open **Candy_Properties.gd** and add a path to the character nodes in your game by editing the **bubbles_npc_path** and **bubbles_player_path** variables:

```
#^ Actor paths:
#? If actor nodes change paths during the course of your game, update dynamicaly.
#? E.g. make your game update these paths during combat, exploration, cutscenes, etc.
var bubbles_npc_path = "/root/Game/Actors"          #/ Path to NPC actor nodes, for speech bubbles.
var bubbles_player_path = "/root/Game/Actors"       #/ Path Player actor node, for speech bubbles.
```

Since NPCs and player-controlled characters often don't have the same paths, different variables are provided for each.

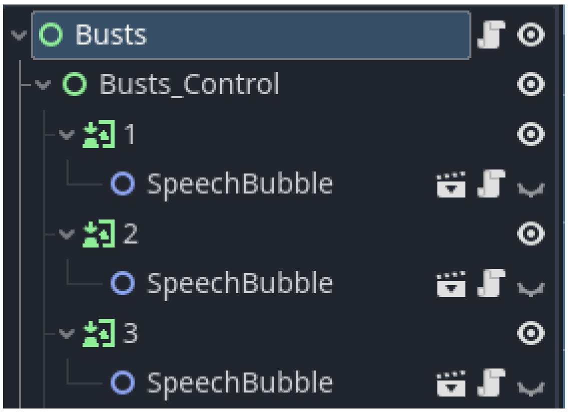
If paths change during the game (e.g. exploration vs. combat vs. cutscenes), you can make your game update these paths dynamically:

1) Create a dictionary to store all possible paths to character nodes in your game.
2) Write a small function to replace the paths in **Candy_Properties.gd** with paths from your dictionary.
3) Call the function when required (e.g. when the game switches between exploration, combat, cutscenes, or whenever else paths change).

With all this data, Candy DE now knows where to find characters, and where to find speech bubbles inside of them. When a character is speaking, Candy DE can display its speech bubble and write text inside of it.

Visual Novels

If you use Candy DE's built-in visual novel scenes, the location of bubbles is a bit more restrictive. VN scenes contain bust nodes which display character images. Each such node must be assigned a speech bubble as a direct child:



There is no need to indicate where the the bubbles are located, because the script of the VN scene root already contains the location of the bust nodes. Candy DE just reads that location to find the node that matches a given character, and assumes the speech bubble is a child of the node.

Note that for this to work, **vn_mode** must be set to true in **Candy_Settings.gd**, otherwise Candy DE will look for speech bubbles in the regular character paths.

Speech Bubble Mode

To enable speech bubbles, in your Dialogue UI scene (e.g. Candy_UI.tscn), in the Inspector, change **dialogue_mode** to "Bubbles".

You should also define an alternate dialogue mode in **bubble_exempt**, in case a speech bubble can't be displayed for a character (such as if the character has no node on screen).

With just these two variables setup, Candy DE will now display spoken text in speech bubbles.

Overrides

You can configure specific characters or specific spoken lines to override the dialogue engine settings and always/never use the **bubble_exempt** mode.

For characters: open **Candy_Database.gd** and, in the **actors** dictionary, set any actor's **"BubblesOverride"** key to 1 to force them to use speech bubbles, or -1 to forbid them from using speech bubbles (0 to disable this override and let the engine settings determine whether or not to use bubbles). Note: if you are upgrading from Candy DE 1.0 to 1.1, **"BubblesOverride"** replaces **"Bubble Exempt"**.

For specific Spoken Lines: in Candy Dialogue Creator, set the 'Bubble Override' data field to -1, 0 or 1.

Note that the line settings take priority over the actor settings. Also, if you force a line or an actor to use speech bubbles, bubbles won't be used if the speaker is not present on screen: the engine will use the alternate dialogue mode you configured in the engine settings, as normal.

This can be useful, for example, to display text for internal thoughts, sound effects, or other forms of special text in a dialogue box, or if an actor is always meant to be an off-screen narrator.

Variables

The default script assigned to speech bubbles features several exported variables you may want to customize to fit your needs:

y_offset → Adjust this value so the bubble is positioned at the right height above the character. Taller characters require higher values.

x_offset → Offsets the entire bubble horizontally, on the X axis.

z_offset → Offsets the entire bubble horizontally, on the Z axis.

max_width → The maximum width the bubble can reach. If the text is longer, the bubble expands vertically.

text_width_margin → Extra width for the RichTextLabel in case text comes too close to the edges.

enable_text_color → Allows the Speech Bubble scene to override the text color (see **text_color** below)

text_color → A custom color for the text. Requires BBCode. Will be overridden by actor settings and BBCode tags inside the text string.